Juan Miguel Cerda Ordaz

Game Development Engineer | Software Engineer | Technical Lead

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Summary

Game Development Engineer with 7+ years building games in Unity, from solo indie work to leading full engineering teams. Co-directed and led development on Heaven Crawler, winner of the Grand Prize at Supernova Devs Challenge 2025 and showcased at Gamescom. Deep hands-on experience across Unity's gameplay, AI, UI, animation, physics, audio, VFX, and performance toolset. Background also includes C++ engine development, Full-stack development, XR/visionOS, and real-time biomedical AI systems. Comfortable running teams, shipping builds, and keeping a codebase clean as a project scales.

Education

Universidad de Artes Digitales **2018 – 2021**
B.S. in Game Development Engineering *Guadalajara, Mexico*

Experience

Founder, Developer & Tech Lead **2019 – 2026**
Critical Slash — Independent Studio *Mexico*

- Founded the studio and spent 6 years solo-developing Heaven Crawler in Unity, from scratch to a fully playable, publicly demoed game played by thousands, before partnering with another studio and co-leading a full production team to scale it toward release. Still actively maintaining and updating the public demo based on community feedback.

Lead Programmer & Co-Director **2021 – 2026**
Carbon Machina Studio *Mexico*

- Led the engineering team building core gameplay, enemy AI (Behavior Trees), procedural generation, UI, animation, physics, VFX, and editor tooling for Heaven Crawler, running Agile sprints, doing code reviews, and keeping architecture clean and scalable as the team grew.
- Co-directed the game's creative vision, characters, story, and gameplay, as the IP's original creator. Profiled and optimized cross-platform performance. Grand Prize, Supernova Game Devs Challenge 2025 (unanimous); showcased at Gamescom and the Latin American Game Showcase.

Professor of Video Game Prototyping **2022 – 2023**
Universidad de Artes Digitales *Guadalajara, Mexico*

- Taught rapid prototyping and gameplay fundamentals in Unity, helping students go from idea to playable build while learning to write clean, maintainable C# and apply OOP and design patterns.

Mobile & Medical AI Engineer **2022 – 2026**
Bowhead Health & Good Place Games OÜ *Remote*

- Built and shipped Mental Chat AI (Unity) on iOS and Android, plus a separate Apple Vision Pro version, and a TikTok Live integration where the AI agent reads viewer comments and gifts in real time and reacts with contextual animations via client-server communication.
- Built a multi-agent AI system for biomedical research with a real-time 3D Virtual Lab (React/Python/FastAPI), including simulations that model cell behavior, growth patterns, and cancer progression.

Projects

Heaven Crawler | *Unity, C#, Behavior Trees, Procedural Gen, Inverse Kinematics, URP* | Lead Programmer & Co-Director**2019 –**

- Action-adventure roguelite dungeon crawler. Grand Prize, Supernova Devs Challenge 2025 (unanimous); Most Anticipated Game Nominee, Game Effect Awards 2025; Official Indie Showcase, Mexican Entertainment System 2025–2026; Gamescom & Latin American Game Showcase. Solo-developed 6 years, then co-led a full production team to commercial release. Active public Steam demo with organic traction from streamers, content creators, and game press.
- Built the enemy AI using Behavior Trees, giving each creature distinct attack patterns and movement states; designed multi-phase Colossal Boss encounters with their own state machines and cinematic transitions; implemented a reactive elemental combat system (fire, ice, electricity, corrosion) that drives dynamic environmental interactions.
- Procedurally generated dungeon layouts across biomes, each with unique room logic, traps, and challenge structures; built a full IK character animation rig; engineered a modular gear system (mix-and-match arms, masks, skills, relics) and the roguelite meta-progression loop (soul carry, lobby upgrades, research unlocks).

Mecha K.O. | *Unity, C#, URP, Toon Shading, Procedural Animation, Cross-Platform* | Lead Gameplay Programmer **2024**

- Narrative-driven 3D action boxing game blending visceral real-time combat with story: a branching campaign with dialogue trees and multiple-choice conversations between fights, giving the fighting game a visual-novel-style narrative layer.
- Mechanical arms physically break apart mid-combat — each arm has 3 structural segments that crack and detach under damage, and the real-time destructibility you see *is* the health system. Built the toon-shaded (cel shading) rendering pipeline in URP, procedural character animation, the structural destruction system, the physics-driven combat model, and the cross-platform Android port solo.

Mental Chat AI | *Unity, C#, iOS, Android, visionOS, AI Chat API* | Lead Programmer **2025 – 2026**

- Shipped on iOS and Android: an AI mental health companion with interactive 3D avatars featuring real-time facial animations and lip-sync that react dynamically to conversation. Separate visionOS build for Apple Vision Pro with spatial UI.
- Built a TikTok Live integration where the AI reads live viewer comments in real time, generates contextual responses, and triggers unique avatar animations and reactions based on viewer gifts and events.

Chrimp Desktop Pet | *Unity, C#, Win32 API, URP, Unity Behavior, Inverse Kinematics* | Solo Developer **2026**

- A desktop pet that lives on top of your screen, transparent, always-on-top, click-through when you're working, built using the Win32 API. Has five minigames each with its own AI: Tower Defense, Football Match, Hot Potato Bomb, Balloon Keep-Up, and Radio Dance (reacts to your music's audio spectrum in real time). Full IK animation rig and a rarity-based species collection system. Coming to Steam.

Biogame AI | *React, Three.js, Python, FastAPI, LangGraph, Neo4j, ChromaDB, Docker* | Sole Architect **2024 – 2025**

- Multi-agent AI bioinformatics SaaS with an isometric 3D virtual lab in Three.js. Built end-to-end: LangGraph orchestration, 12+ biomedical tool modules, live database integrations (PubMed, ClinicalTrials.gov, UniProt), Neo4j knowledge graph, Stripe billing, deployed on Google Cloud Run.

Vivarium Engine | *C++, OpenGL, DirectX, Nvidia Omniverse API, PhysX* | Solo Developer **2020 – 2021**

- Custom C++ 3D engine with deferred PBR rendering, PhysX, VR support, NavMesh baking, and a live Nvidia Omniverse connector built while the platform was still in early development — one of the first independent integrations. Invited to present at Nvidia GTC 2021 and featured on the Nvidia developer blog.

Other Unity Projects: Ramen Abyss *(targeting Android/iOS)*, American Trail, Cyberdogz *(Photon Networking)*, Slug Slayer, Skate & Slash **Other:** Docs Slayer *(live ops)*, Doppelbio, Minicule *(live ops)*, Propermax *(live ops)*, Exact Agent *(live ops)*, SOLBAG

Technical Skills

Languages: C#, C++, Python, JavaScript, TypeScript, HTML5, CSS3
Platforms: iOS, Android, PC (Steam), VR/XR (visionOS, Apple Vision Pro)
Engines & Tools: Unity, Unreal Engine, React, Three.js, FastAPI, Node.js, Docker, Git, Unity Profiler, Xcode Instruments, Photon, FMOD
Specialties: Core Gameplay Engineering, Enemy AI & Behavior Trees, Procedural Generation, Inverse Kinematics, Gamefeel & Polish, OOP & Design Patterns, Mobile Performance Optimization, Agile/Scrum